1 Training-free inpainting with a diffusion prior

1.1 Diffusion prior and notation

All inpainting algorithms considered in this work operate on a single pre-trained denoising diffusion probabilistic model (DDPM) [1] per centrality class, used *as is*: no component is retrained or fine-tuned for the inpainting task. The model acts in the normalized (logarithmic) representation of the calorimeter image, $x = \ln\big(\max(E_T, E_{\min})\big)$ with $E_{\min} = 10^{-3}$ GeV, on images $x \in \mathbb{R}^{1\times 24\times 64}$ in the (η, ϕ) plane.

The forward process is the standard variance-preserving Markov chain with T steps,

$$q(x_t \mid x_{t-1}) = \mathcal{N}\big(x_t;\ \sqrt{1-\beta_t}\, x_{t-1},\ \beta_t \mathbf{I}\big), \qquad q(x_t \mid x_0) = \mathcal{N}\big(x_t;\ \sqrt{\bar\alpha_t}\, x_0,\ (1-\bar\alpha_t)\,\mathbf{I}\big), \tag{1}$$

with $\alpha_t = 1 - \beta_t$ and $\bar\alpha_t = \prod_{i\le t} \alpha_i$. The schedule replicates the training configuration exactly: $T = 8000$ and a linear schedule β_t interpolating from $\beta_1 = b/T$ to $\beta_T = 1000\, b/T$, with $b = 0.02$ (0.10) for the 0–10% (40–50%) centrality model. The network is an ϵ-prediction U-Net, $\hat\epsilon = \epsilon_\theta(x_t, t)$, from which the denoised estimate follows by Tweedie's formula,

$$\hat x_0(x_t) = \frac{x_t - \sqrt{1-\bar\alpha_t}\ \epsilon_\theta(x_t, t)}{\sqrt{\bar\alpha_t}}. \tag{2}$$

For sampling efficiency the reverse process is run on a subsampled time grid $0 = \tau_0 < \tau_1 = 1 < \cdots < \tau_S = T$ obtained by rounding a linear spacing of S points ($S = 1000$ unless stated otherwise); the network is always conditioned on the *original* time indices τ_k. Writing $t = \tau_k$ and $s = \tau_{k-1}$ for one reverse step, the effective per-step noise level is $\beta_{(k)} \equiv 1 - \bar\alpha_t/\bar\alpha_s$, and the DDPM posterior used by the ancestral sampler is

$$q(x_s \mid x_t, x_0) = \mathcal{N}\left(x_s;\ \frac{\sqrt{\bar\alpha_s}\,\beta_{(k)}}{1-\bar\alpha_t}\, x_0 + \frac{\sqrt{\bar\alpha_t/\bar\alpha_s}\,(1-\bar\alpha_s)}{1-\bar\alpha_t}\, x_t,\ \tilde\beta_{(k)}\mathbf{I}\right), \qquad \tilde\beta_{(k)} = \frac{(1-\bar\alpha_s)\,\beta_{(k)}}{1-\bar\alpha_t}. \tag{3}$$

Because $\bar\alpha_{\tau_0} = \bar\alpha_0 = 1$, the final reverse step collapses, $q(x_0 \mid x_{\tau_1}, \hat x_0) = \delta(x_0 - \hat x_0)$; this property is used below to guarantee exact data consistency of the final sample. Unconditional generation initializes $x_T \sim \mathcal{N}(0, (1-\bar\alpha_T)\mathbf{I})$ and iterates Eqs. (2)–(3) with $x_0 \to \hat x_0$.

1.2 Noise-free inpainting setting

A dead detector region is described by a binary mask $m \in \{0,1\}^{24\times 64}$, with $m_i = 1$ on live (observed) towers and $m_i = 0$ on dead towers. The measurement is noise-free,

$$y = m \odot x_0^{\text{true}}, \qquad \sigma_y = 0, \tag{4}$$

i.e. the degradation operator is $A = \mathrm{diag}(m)$ and the inference target is the posterior $p\big(x_0 \mid m\odot x_0 = y\big)$ under the DDPM prior. Two simplifications specific to this setting are used throughout. First, A is its own Moore–Penrose pseudoinverse on its range, $A^\dagger = A^\top = A$, so no singular-value machinery is required: spectral space coincides with pixel space, with unit singular values on observed and zero on dead pixels. Second, all branches of the algorithms associated with measurement noise ($\sigma_y > 0$) drop out identically. In this noiseless-masking limit the five algorithms below—although derived from different principles (heuristic projection, null-space decomposition, spectral variational inference, and two flavors of guidance by the measurement-consistency gradient)—approximate the *same* posterior, which makes their head-to-head calibration comparison meaningful.

Posterior sampling is performed by batching n independent reverse trajectories per image ($n = 50$ in the main study); each algorithm consumes one measurement pair (y, m) and returns n samples of the full image, of which only the dead region is retained for analysis.

1.3 Algorithms

Throughout this section one reverse step from $t = \tau_k$ to $s = \tau_{k-1}$ is described; $\xi, \xi', \dots \sim \mathcal{N}(0, \mathbf{I})$ denote independent standard normal draws, and $\hat{x}_0 \equiv \hat{x}_0(x_t)$, $\hat{\epsilon} \equiv \epsilon_\theta(x_t, \tau_k)$ from Eq. (2).

1.3.1 RePaint

RePaint [2] alternates an unconditional reverse step on the dead region with an independent forward-diffusion of the known content to the *matching* noise level:

$$x_s^{\text{known}} \sim q(x_s \mid x_0 = y) = \mathcal{N}\big(\sqrt{\bar{\alpha}_s}\, y,\ (1 - \bar{\alpha}_s)\,\mathbf{I}\big), \tag{5}$$

$$x_s^{\text{dead}} \sim q(x_s \mid x_t, \hat{x}_0) \quad \text{[Eq. (3)]}, \tag{6}$$

$$x_s = m \odot x_s^{\text{known}} + (1 - m) \odot x_s^{\text{dead}}. \tag{7}$$

We emphasize the time index in Eq. (5): both branches must live at level $s = \tau_{k-1}$ *after* the step. Noising the known region to level τ_k instead—a plausible-looking off-by-one— leaves it systematically one step too noisy at every iteration and biases the reconstruction at the dead-region boundary; we verified that this variant degrades calibration and use the correct τ_{k-1} convention.

To harmonize the two branches, RePaint applies *resampling*: steps (5)–(7) are repeated U times ($U = 10$ by default), jumping back one step between repetitions with the forward kernel $x_t \sim q(x_t \mid x_s) = \mathcal{N}(\sqrt{\bar{\alpha}_t / \bar{\alpha}_s}\, x_s,\ \beta_{(k)} \mathbf{I})$.[1] At the final step $\bar{\alpha}_{\tau_0} = 1$, so Eq. (5) returns y exactly: the known region of the output equals the measurement identically.

1.3.2 DDNM

The Denoising Diffusion Null-space Model [3] rectifies the range-space component of the denoised estimate at every step. Any solution consistent with $y = Ax_0$ can be written $x_0 = A^\dagger y + (\mathbf{I} - A^\dagger A)\,\bar{x}$; for noiseless masking the projection of $\hat{x}_0$ onto this affine subspace is simply

$$\hat{x}_0^{\text{proj}} = m \odot y + (1 - m) \odot \hat{x}_0, \tag{8}$$

after which the standard ancestral step, Eq. (3) with $x_0 \to \hat{x}_0^{\text{proj}}$, is taken. No noisy-case corrections (the Σ-scaling of DDNM$^+$) are needed for $\sigma_y = 0$. By the collapse property of the final step, the output equals $\hat{x}_0^{\text{proj}}$ at $k = 1$, so data consistency of the known region is again exact.

1.3.3 DDRM

Denoising Diffusion Restoration Models [4] define a variational posterior in the spectral domain of A. For masking the spectral basis is the pixel basis, and with $\sigma_y = 0$ the general update [Eqs. (7)–(8) of Ref. [4]] reduces, in the variance-preserving variables used here, to

$$\text{dead pixels:} \quad x_s = \sqrt{\bar{\alpha}_s}\,\hat{x}_0 + \sqrt{(1 - \eta^2)(1 - \bar{\alpha}_s)}\,\hat{\epsilon} + \eta\sqrt{1 - \bar{\alpha}_s}\,\xi, \tag{9}$$

$$\text{observed pixels:} \quad x_s = \sqrt{\bar{\alpha}_s}\,\big[(1 - \eta_b)\,\hat{x}_0 + \eta_b\, y\big] + \sqrt{1 - \bar{\alpha}_s}\,\xi', \tag{10}$$

with hyperparameters $\eta = 0.85$ and $\eta_b = 1$ at their published defaults. Initialization follows the same pattern at $t = T$: observed pixels start from $\mathcal{N}(\sqrt{\bar{\alpha}_T}\, y, (1 - \bar{\alpha}_T)\mathbf{I})$ and dead pixels from $\mathcal{N}(0, (1 - \bar{\alpha}_T)\mathbf{I})$. Note that Eq. (9) is DDIM-like but references the *marginal* noise scale $\sqrt{1 - \bar{\alpha}_s}$ rather than the ancestral posterior width $\tilde{\beta}_{(k)}^{1/2}$—a DDRM-specific choice. With $\eta_b = 1$ the observed pixels are independently re-noised copies of y at every level, and at the final step ($\bar{\alpha}_0 = 1$, vanishing noise) the output known region equals y exactly while dead pixels return $\hat{x}_0$.

[1] Algorithm 1 of Ref. [2] writes the jump-back kernel with β_{t-1}; the kernel consistent with the forward process (1) uses the transition variance of the step being re-traversed, which is what we implement.

1.3.4 MCG

Manifold-constrained gradients [5] augment the RePaint-style projection with a measurement-consistency gradient taken *through the denoiser network*. With $\mathcal{R}(x_t) = \big\|m \odot \big(y - \hat{x}_0(x_t)\big)\big\|_2$ (the ℓ_2 norm, not its square, following the official implementation—this makes the correction self-normalizing), one step reads

$$g = \nabla_{x_t} \mathcal{R}(x_t), \tag{11}$$

$$x_s^{\text{dead}} = x_s' - \alpha\, g, \qquad x_s' \sim q(x_s \mid x_t, \hat{x}_0), \tag{12}$$

$$x_s = m \odot x_s^{\text{known}} + (1-m) \odot x_s^{\text{dead}}, \tag{13}$$

with x_s^{known} as in Eq. (5) (same τ_{k-1} convention) and step size $\alpha = 1$. Note the division of roles: the residual $\mathcal{R}$ is evaluated on the *observed* pixels, while the correction acts on the *dead* pixels—the two regions being coupled through the network Jacobian, which is what allows observed-data consistency to steer the reconstruction inside the dead region (the known pixels themselves are overwritten by the projection). The gradient requires one backward pass through ϵ_θ per step; the prior network weights remain frozen, and differentiation is with respect to x_t only. Exact data consistency of the known region follows from the projection as for RePaint.

1.3.5 ΠGDM

Pseudoinverse-guided diffusion models [6] approximate the intractable guidance term $\nabla_{x_t} \log p_t(y \mid x_t)$ using a Gaussian surrogate for $p(x_0 \mid x_t)$. For a noiseless measurement the guidance reduces to a vector–Jacobian product with the mask-projected residual [Eq. (8) of Ref. [6]; the r_t^2 prefactor cancels exactly in the update for $\sigma_y = 0$],

$$g = \left(\frac{\partial \hat{x}_0(x_t)}{\partial x_t}\right)^{\top} \big[\, m \odot (y - \hat{x}_0)\,\big], \tag{14}$$

evaluated with one backward pass through the network (the residual is treated as a constant cotangent). The update is a DDIM(η) step [7] plus the guidance scaled by $\sqrt{\bar{\alpha}_t}$, as required for variance-preserving parametrizations [Algorithm 1 of Ref. [6]]:

$$x_s = \sqrt{\bar{\alpha}_s}\, \hat{x}_0 + c_2\, \hat{\epsilon} + c_1\, \xi + \sqrt{\bar{\alpha}_t}\, g, \qquad c_1 = \eta\, \tilde{\beta}_{(k)}^{1/2}, \quad c_2 = \sqrt{1 - \bar{\alpha}_s - c_1^2}, \tag{15}$$

with $\eta = 1$ (for which the first three terms are equal in law to the ancestral step). Following the convention of published implementations (`clip_denoised`), $\hat{x}_0$ is clamped to the physical data range in log space before entering the update and the guidance cotangent; this is essential rather than cosmetic. The guidance forms a feedback loop ($\hat{x}_0 \to$ residual $\to$ vector–Jacobian product $\to x_t \to \hat{x}_0$) that is bounded only by the near-cancellation $(\mathbf{I} - \sqrt{1-\bar{\alpha}_t}\, \partial\hat{\epsilon}/\partial x_t)/\sqrt{\bar{\alpha}_t}$; with $1/\sqrt{\bar{\alpha}_t}$ reaching ~ 150 on this schedule, the trained network's residual Jacobian imperfection is amplified multiplicatively and the unclamped chain diverges within a few steps (in single precision as well). The clamp bounds the loop and, having zero gradient in saturation, switches the guidance off exactly where $\hat{x}_0$ leaves the physical range. Unlike the other four algorithms, ΠGDM enforces consistency during sampling only through the guidance term, so the known region of its raw output matches y approximately rather than identically. For the noise-free setting we project the known region to y at the final step; since this occurs after the last network evaluation, it is pure post-processing—it cannot affect the sampling dynamics or the dead-region pixels on which the analysis is based.

Table 1: Summary of the implemented inpainting algorithms, specialized to noise-free masking. "Backprop" indicates a backward pass through the frozen ϵ-network per reverse step; "exact y" indicates that the known region of the final sample equals the measurement identically. NFEs are network function evaluations per posterior sample for S reverse steps. (†: via a final-step projection applied after the last network evaluation; the published algorithm is only approximately consistent.)

Algorithm	Hyperparameters	Backprop	Exact y	NFEs
RePaint [2]	$U = 10$	no	yes	$U(S-1)+1$
DDNM [3]	—	no	yes	S
DDRM [4]	$\eta = 0.85$, $\eta_b = 1$	no	yes	S
MCG [5]	$\alpha = 1$	yes	yes	S (+S bwd)
ΠGDM [6]	$\eta = 1$	yes	yes†	S (+S bwd)

1.4 Implementation and validation

All five algorithms share the sampler infrastructure: the exact training schedule reconstructed from the model configuration, the subsampled time grid of Sec. 1.1, evaluation of ϵ_θ under `bfloat16` autocast with all schedule and update arithmetic in `float32`, and a dedicated counter-based random stream per truth image (making every study run bitwise reproducible and restartable). Truth images are drawn from the same DDPM used as the prior, so the ground truth is, by construction, a sample from the prior—the setting in which posterior-calibration diagnostics cleanly isolate algorithmic approximation error. Table 1 summarizes the algorithm-specific choices.

The implementation is validated in two independent ways. First, the schedule, subsampling, and posterior coefficients are verified numerically against the training code of Ref. [8] to single precision. Second, each algorithm is run on an analytically tractable problem: for an i.i.d. Gaussian prior $x_0 \sim \mathcal{N}(\mu_0, \sigma_0^2 \mathbf{I})$ the optimal ϵ-predictor is known in closed form and the true posterior on dead pixels is again $\mathcal{N}(\mu_0, \sigma_0^2)$, independent of the observed pixels. RePaint, DDNM, MCG, and ΠGDM reproduce this posterior to within sampling accuracy, and the algorithms with hard projection preserve the known region exactly (to floating-point precision). DDRM at its default $\eta = 0.85$ reproduces the posterior mean but *under-disperses* the posterior (standard-deviation ratio ≈ 0.86 in this test, approaching 1 as $\eta \to 0$ and ≈ 0.70 at $\eta = 1$, independent of S)—an intrinsic property of its variational posterior in the noise-free limit rather than an implementation artifact, and one that the calibration study of Sec. 3 is designed to expose.

2 Statistical validation methodology

2.1 Evaluation frame

Truth images are drawn from the same DDPM that serves as the inpainting prior. The ground truth is therefore, by construction, a sample from the prior, and for a noiseless measurement the pair $(x_0^{\text{true}},\, y = m \odot x_0^{\text{true}})$ realizes exactly the self-consistency setting of simulation-based calibration [9]: if an algorithm sampled the posterior $p(x_0 \mid y)$ exactly, truth and posterior samples would be exchangeable, and any measurable departure from the calibration diagnostics below is attributable purely to algorithmic approximation error (there is no model misspecification by construction).

For each truth image $i = 1, \dots, N$ the algorithm under test draws J i.i.d. posterior samples $\{x^{(i,j)}\}_{j=1}^{J}$ ($J = 50$, $N = 1000$ in the main study); all diagnostics are evaluated on the dead-region pixels only. Perceptual or point-estimate image metrics (FID, LPIPS, PSNR, SSIM, Pearson correlation of the posterior mean) are deliberately not used: the scientific target is the posterior *distribution* over physical quantities, not the fidelity of a single reconstruction. In particular, the posterior mean is an intentionally variance-suppressed point estimate; regressing it against truth necessarily yields a slope below unity. This shrinkage slope is reported as a diagnostic of the posterior width but is not scored.

2.2 Per-pixel rank statistics (SBC)

For every image i and dead pixel p, the rank of truth among the posterior samples is

$$r_{ip} \;=\; \#\big\{\, j : x_p^{(i,j)} < x_{ip}^{\text{true}} \big\} \;+\; u_{ip}, \qquad u_{ip} \sim \mathcal{U}\{0,\ldots,\#\{j : x_p^{(i,j)} = x_{ip}^{\text{true}}\}\}, \tag{16}$$

where the randomized tie-breaking term u_{ip} handles exact ties (which occur at the $E_{\min}$ clipping floor). Under exact calibration r_{ip} is uniform on $\{0,\ldots,J\}$ [9]. Ranks are invariant under strictly monotone transformations, so the GeV and log-space analyses coincide. We report the pooled rank histogram with a χ^2 uniformity p-value and, separately, per-pixel p-value maps across the dead-region box. The histogram shape identifies the failure mode: a U-shape indicates an under-dispersed posterior (truth falls in the tails too often), a central hump over-dispersion, and a monotone slope a systematic bias.

2.3 Joint coverage (TARP)

Per-pixel ranks are insensitive to miscalibration of the *correlations* between dead pixels. Joint calibration over the $d = b^2$ box pixels is tested with the expected-coverage-probability estimator of Ref. [10] (TARP). Coordinates are standardized in log space using the pooled per-dimension mean and standard deviation of all posterior samples. For each image a random reference point $\theta_r^{(i)}$ is drawn uniformly over the pooled per-dimension range of samples and truths, and

$$f_i \;=\; \frac{1}{J}\sum_{j=1}^{J} \mathbf{1}\Big[\, \big\|x^{(i,j)} - \theta_r^{(i)}\big\|_2 \;<\; \big\|x_i^{\text{true}} - \theta_r^{(i)}\big\|_2 \Big]. \tag{17}$$

Under exact calibration $f_i \sim \mathcal{U}(0,1)$, equivalently the expected coverage $\mathrm{ECP}(\alpha) = \Pr(f_i \le \alpha)$ lies on the diagonal. We report the ECP curve, its maximum absolute deviation from the diagonal, and a Kolmogorov–Smirnov p-value for $f \sim \mathcal{U}(0,1)$. Under-dispersion appears as an ECP curve sagging below the diagonal.

2.4 Pulls and central credible intervals

Two complementary, more interpretable diagnostics are evaluated in log space. The per-pixel pull

$$z_{ip} = \frac{x_{ip}^{\text{true}} - \widehat{\mu}_{ip}}{\widehat{\sigma}_{ip}}, \qquad \widehat{\mu}_{ip} = \tfrac{1}{J}\textstyle\sum_j x_p^{(i,j)}, \quad \widehat{\sigma}_{ip}^2 = \tfrac{1}{J-1}\textstyle\sum_j \big(x_p^{(i,j)} - \widehat{\mu}_{ip}\big)^2, \tag{18}$$

is approximately standard normal for a calibrated, locally near-Gaussian posterior; the pooled pull mean measures bias and the pooled pull width the dispersion ratio (width $> 1 \Leftrightarrow$ under-dispersed posterior). Independently, central credible intervals at nominal levels $\gamma = 0.1,\ldots,0.9$ are formed from the empirical $\big(\frac{1\mp\gamma}{2}\big)$-quantiles of the J samples, and empirical coverage is compared to γ both pooled over dead pixels and for the physics observable $\Sigma E = \sum_{p\in\text{box}} E_p$ (in GeV), the latter probing a physically relevant joint functional of the posterior.

2.5 Comparison protocol and robustness

All algorithms are evaluated on identical truth images, masks, and per-image random streams, so cross-algorithm differences are paired. The comparison figure overlays, per algorithm, the central-coverage curves and the deviation of the empirical rank CDF from uniformity. Statistical power at the study scale: pooled rank tests use Nb^2 ranks across $J+1$ bins; the TARP ECP has standard error $\sim \sqrt{\alpha(1-\alpha)/N}$, so $N = 1000$ resolves coverage distortions at the few-percent level. Analysis-level robustness safeguards: images containing non-finite samples are excluded and their count reported (a nonzero count indicates a sampler failure, not an analysis artifact), and partially completed runs are analyzed up to their checkpoint, enabling monitoring of running studies.

2.6 Generation-stage verification

Upstream of the inpainting study, the event-generation stage is verified against Ref. [8]: tower-area E_T spectra (1×1, 4×4, 7×7, 11×11, sliding windows, periodic in ϕ) and the ΣE_T distribution per centrality, overlaid across the five independently trained model seeds with seed-ratio panels; the event-averaged $\langle \sigma_{E_T} \rangle$ versus ΣE_T profiles; regression anchors on $\langle \Sigma E_T \rangle$ per centrality (soft ±5% checks against validated reference runs); and a monitor of the high-energy single-tower tail reporting the rate of towers above 10 GeV per seed (towers of $\mathcal{O}$(10–20) GeV are kinematically plausible at $\sqrt{s_{NN}} = 200$ GeV; only energies beyond 50 GeV are treated as unphysical).

3 Results

(Placeholder — in the paper this is the calibration-study section.)

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